# Overview of Jamming Communications and Future Plans

## Jamming Communications

Student ID: 1280391
Name: Natsuka Hosokawa

Kochi University of Technology

June 3, 2026

# Progress

1. Jamming in SAGIN
2. Overview of SAGIN
3. Overview of Jamming (Current)

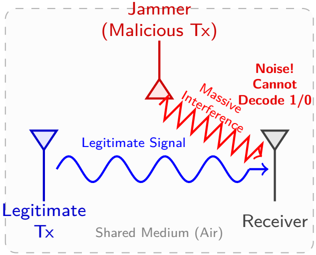
# Table of Contents

## What is Jamming?

# Overview

Since wireless communication relies on a shared medium, jamming is relatively easy to execute and difficult to defend against.

If the interference (noise) is too high, the receiver cannot distinguish between 1 and 0, making communication impossible.




## Summary

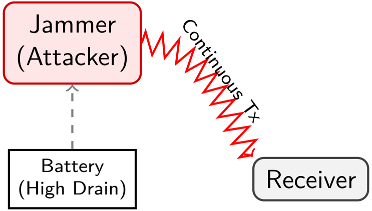
Jamming is an attack that intentionally broadcasts high-power interference signals into a shared medium to drown out legitimate communication.

# What is Jamming?

# Types of Jamming

- **Continuous Jamming**:
  Continuously emits high-power interference signals.

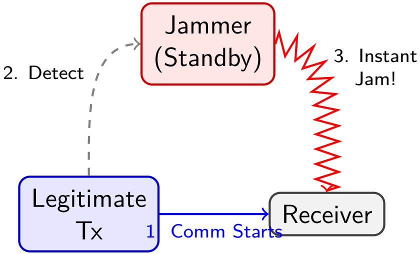
  Very effective at disrupting communication, but rapidly depletes the attacker's battery.



# Types of Jamming

- **Reactive Jamming**:
  Remains idle normally, but transmits an interference signal the moment legitimate communication is detected.

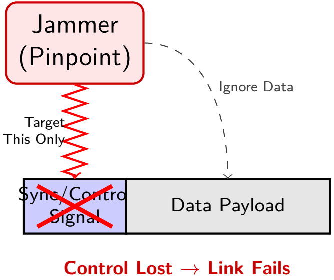
  Highly energy-efficient and difficult to detect since it only operates when necessary.



# Types of Jamming

- **Smart Jamming**:
  Rather than jamming all data, it targets critical control packets or synchronization signals required to establish a connection.

  Commonly used against `Wi-Fi` and 5G networks.



# Summary

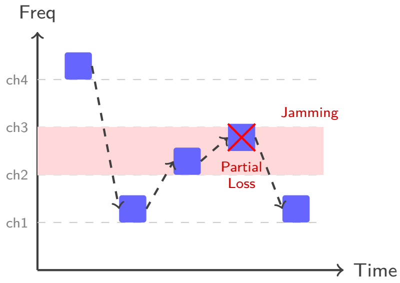
As outlined above, there are various types of jamming attacks, each with distinct advantages, disadvantages, and specific use cases depending on the scenario.

# Anti-Jamming Countermeasures

# Common Countermeasures

- **Frequency Hopping**:
  Rapidly switches the carrier frequency during transmission based on a predetermined pseudorandom sequence.

  Makes it extremely difficult for an attacker to predict and jam the active frequency band. Widely used in `Bluetooth`.

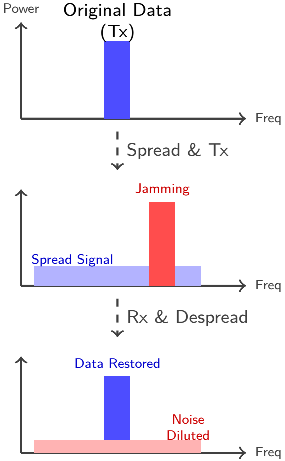



**Rapid Switching SECURES Comm**

# Common Countermeasures

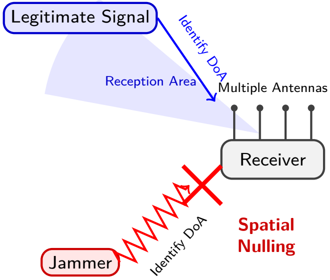
- **Spread Spectrum**:
  A method that spreads the transmitted signal over a much wider frequency band than originally required.

  Highly resilient to noise. When the receiver despreads the signal, the jammer's power is diluted across the wideband, neutralizing the attack.



- `MIMO` **and Spatial Filtering**:
  Utilizes multiple antennas at both the base station and the user equipment.

  By leveraging multiple antennas, the receiver can calculate the Direction of Arrival (DoA) of both the legitimate signal and the jamming signal. The receiver can then create a spatial "null" to physically block the interference.



# Anti-Jamming Countermeasures

# The Nature of Security

## An Endless Cat-and-Mouse Game

Naturally, these countermeasures do not provide perfect protection. Attackers continuously develop advanced methods (counter-countermeasures) to bypass these defenses. This evolutionary arms race is the fundamental nature of cybersecurity. While it is a complex field, I plan to delve into these advanced adversarial dynamics in my future studies.

# Anti-Jamming Countermeasures

# Summary

While various anti-jamming techniques exist, none offer a universal shield against all attack vectors. It is crucial to understand how defenses are implemented, how attackers attempt to circumvent them, and to continually devise robust, next-generation countermeasures.

# Current Progress and Future Plans

# Simulation Environment

Setup for Jamming Communications Simulation

- Server Environment (TBD):
  However, `Dockerfile` and `docker-compose.yaml` have already been created, enabling immediate deployment.
- Required Library (**`PyJama`**):
  - Expected to run on `Python 3.10-slim`.
  - Assumes the use of a `GPU` to accelerate matrix operations and tensor calculations.
  - Available as `OSS` on GitHub: `https://github.com/IIP-Group/pyjama`

# Current Progress and Future Plans

# Future Study Plans

- Study the underlying mathematics and mechanisms of signal jamming.
- Study Reinforcement Learning (RL) in Prof. Wei's seminar.
- Study Game Theory to predict optimal jamming and defense strategies, which will be integrated with RL models.

- Develop a simplified `Python` program to simulate basic jamming attacks.

# References

# References I


Hossein Pirayesh and Huacheng Zeng.
Jamming Attacks and Anti-Jamming Strategies in Wireless Networks: A Comprehensive Survey.
*IEEE Communications Surveys & Tutorials*, 2022.

Jaewon Yun, Joohyuk Park, and Yo-Seb Jeon.
Anti-Jamming Modulation for OFDM Systems under Jamming Attacks.
*arXiv preprint arXiv:2510.02640*.

Michal Harvanek, et al.
Survey on 5G Physical Layer Security Threats and Countermeasures.
*Sensors*, 2024.

# References II

Takaki Amagai.
On Next-Generation Electronic Warfare: EMS Operations Utilizing Machine Learning and Networks.
*Strategic Studies of JMSDF Staff College*, 2020.